

glmCH2

Load and clean data

```
#load packages  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr   1.5.1  
## v ggplot2    3.5.2      v tibble    3.3.0  
## v lubridate  1.9.4      v tidyr     1.3.1  
## v purrr      1.0.4  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)  
library(DHARMa)
```

```
## This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type news(package = 'DHARMa')
```

```
library(glmmTMB)  
library(ggeffects)  
library(car)
```

```
## Loading required package: carData  
##  
## Attaching package: 'car'  
##  
## The following object is masked from 'package:dplyr':  
##  
##      recode  
##  
## The following object is masked from 'package:purrr':  
##  
##      some
```

```
#read in data  
data <- read.csv("06_cleandata/combined_plot_data.csv")  
  
#change character columns to factors
```

```
surveys <- data %>%
  mutate(
    site_id = as.factor(site_id),
    round = as.factor(round),
    plot_id = as.factor(plot_id),
    bee_date = as.Date(bee_date),
    bee_season_type = as.factor(bee_season_type),
    area = as.factor(area))

str(surveys)
```

```
## 'data.frame':    210 obs. of  34 variables:
## $ site_id      : Factor w/ 12 levels "avtar_168","bates_57b",...: 2 2 2 2 2 2 2 2 2 2 ...
## $ round       : Factor w/ 6 levels "2024_r1","2024_r2",...: 1 1 1 2 2 2 3 3 3 4 ...
## $ plot_id     : Factor w/ 3 levels "d","p1","p2": 2 3 1 2 3 1 2 3 1 2 ...
## $ bee_date    : Date, format: "2024-04-15" "2024-04-15" ...
## $ julian.date : int  106 106 106 158 158 158 173 173 173 101 ...
## $ bee_season_type : Factor w/ 2 levels "nestsearching",...: 1 1 1 2 2 2 2 2 2 1 ...
## $ area        : Factor w/ 6 levels "delta_1","delta_2",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ cover_type   : chr  "grass" "grass" "grass" "grass" ...
## $ bare_perc    : num  2 1.42 0 11.92 20.5 ...
## $ grass_perc   : num  46.8 52.2 90.4 52.7 56.8 ...
## $ forb_perc    : num  9.33 7.92 8 16.75 8.12 ...
## $ grass_forb_perc : num  56.2 60.2 98.4 69.4 64.9 ...
## $ grass_h      : num  17.9 19.5 28.8 25.3 25.7 ...
## $ forb_h       : num  9.83 7.42 9.58 26.17 17.42 ...
## $ floral_perc  : num  0.375 1.0417 0.125 0.0417 0 ...
## $ bombus_floral_perc : num  0.375 1.0417 0.0833 0.0417 0 ...
## $ grass_forb_floral_perc : num  56.5 61.2 98.5 69.5 64.9 ...
## $ flood_perc   : num  0 0 0 0 0 0 0 0 0 0 ...
## $ dry_root_perc : num  25.42 37.17 7.08 16.25 5.42 ...
## $ bryo_perc    : num  0 0 0 0 0 0 0 0 0 0 ...
## $ bare_flood_dry_bryo_perc: num  27.42 38.58 7.08 28.17 25.92 ...
## $ bare_dry_bryo_perc : num  27.42 38.58 7.08 28.17 25.92 ...
## $ floral_richness : int  1 1 3 2 3 14 2 2 12 2 ...
## $ bombus_floral_richness : int  1 1 2 2 3 11 2 2 10 2 ...
## $ floral_abun    : num  3 3 3.32 2.01 2.01 ...
## $ bombus_floral_abun : num  3 3 3.04 2.01 2.01 ...
## $ nest_searching_queens : int  3 0 4 2 0 0 0 0 0 0 ...
## $ workers        : int  1 0 0 0 1 2 0 0 0 0 ...
## $ forage_vaco     : int  1 2 0 2 4 0 0 0 0 3 ...
## $ num_burrows     : int  41 120 93 41 120 93 41 120 93 41 ...
## $ num_inactive_burrows : int  10 64 0 10 64 0 10 64 0 10 ...
## $ num_active_dm   : int  5 6 0 5 6 0 5 6 0 5 ...
## $ num_dm_size     : int  3 3 0 3 3 0 3 3 0 3 ...
## $ num_inactive_dm : int  1 2 0 1 2 0 1 2 0 1 ...
```

Add plot type column (ditch vs field)

```
surveys <- surveys %>%
  mutate(plot_type = if_else(plot_id == "d", "ditch", "field"))

surveys <- surveys %>%
```

```
mutate(
  plot_type = as.factor(plot_type))
```

Scale continuous predictors

```
surveys$julian.date_scaled <- as.numeric(scale(surveys$julian.date))
surveys$bare_perc_scaled <- as.numeric(scale(surveys$bare_perc))
surveys$grass_perc_scaled <- as.numeric(scale(surveys$grass_perc))
surveys$forb_perc_scaled <- as.numeric(scale(surveys$forb_perc))
surveys$grass_forb_perc_scaled <- as.numeric(scale(surveys$grass_forb_perc))
surveys$grass_h_scaled <- as.numeric(scale(surveys$grass_h))
surveys$forb_h_scaled <- as.numeric(scale(surveys$forb_h))
surveys$floral_perc_scaled <- as.numeric(scale(surveys$floral_perc))
surveys$grass_forb_floral_perc_scaled <- as.numeric(scale(surveys$grass_forb_floral_perc))
surveys$bombus_floral_perc_scaled <- as.numeric(scale(surveys$bombus_floral_perc))
surveys$flood_perc_scaled <- as.numeric(scale(surveys$flood_perc))
surveys$dry_root_perc_scaled <- as.numeric(scale(surveys$dry_root_perc))
surveys$bryo_perc_scaled <- as.numeric(scale(surveys$bryo_perc))
surveys$bare_flood_dry_bryo_perc_scaled <- as.numeric(scale(surveys$bare_flood_dry_bryo_perc))
surveys$floral_richness_scaled <- as.numeric(scale(surveys$floral_richness))
surveys$bombus_floral_richness_scaled <- as.numeric(scale(surveys$bombus_floral_richness))
surveys$floral_abun_scaled <- as.numeric(scale(surveys$floral_abun))
surveys$bombus_floral_abun_scaled <- as.numeric(scale(surveys$bombus_floral_abun))
surveys$nest_searching_queens_scaled <- as.numeric(scale(surveys$nest_searching_queens))
surveys$workers_scaled <- as.numeric(scale(surveys$workers))
surveys$forage_vaco_scaled <- as.numeric(scale(surveys$forage_vaco))
surveys$num_burrows_scaled <- as.numeric(scale(surveys$num_burrows))
surveys$num_inactive_burrows_scaled <- as.numeric(scale(surveys$num_inactive_burrows))
surveys$num_active_dm_scaled <- as.numeric(scale(surveys$num_active_dm))
surveys$num_dm_size_scaled <- as.numeric(scale(surveys$num_dm_size))
surveys$num_inactive_dm_scaled <- as.numeric(scale(surveys$num_inactive_dm))
```

GLMs #1: Number of nest-searching queens vs plot type (ditch vs field)

Plotting a zero-inflated poisson model

```
nsq_glm_zeroPois <- glmmTMB(nest_searching_queens ~ plot_type +
  bee_season_type +
  (1|area/site_id/plot_id),
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = surveys)

summary(nsq_glm_zeroPois)
```

```
## Family: poisson ( log )
## Formula:
## nest_searching_queens ~ plot_type + bee_season_type + (1 | area/site_id/plot_id)
## Zero inflation: ~1
```

```

## Data: surveys
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    299.4    322.8   -142.7    285.4      203
##
## Random effects:
##
## Conditional model:
##   Groups          Name      Variance Std.Dev.
## plot_id:site_id:area (Intercept) 1.704e-01 4.128e-01
##   site_id:area      (Intercept) 1.066e-01 3.264e-01
##   area              (Intercept) 3.395e-09 5.826e-05
## Number of obs: 210, groups:
## plot_id:site_id:area, 36; site_id:area, 12; area, 6
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.6268     0.5498   1.140 0.254252
## plot_typefield    -1.1086     0.3074  -3.606 0.000311 ***
## bee_season_typer  -2.4574     0.4157  -5.911 3.4e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.6221     0.8837  -0.704   0.481

##area random effect has zero variance. remove area from model and check summary

nsq_glm_zeroPois_noArea <- glmmTMB(nest_searching_queens ~ plot_type +
  bee_season_type +
  (1|site_id/plot_id),
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = surveys)

summary(nsq_glm_zeroPois_noArea)

## Family: poisson ( log )
## Formula:
## nest_searching_queens ~ plot_type + bee_season_type + (1 | site_id/plot_id)
## Zero inflation: ~1
## Data: surveys
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    297.4    317.5   -142.7    285.4      204
##
## Random effects:
##
## Conditional model:
##   Groups          Name      Variance Std.Dev.
## plot_id:site_id (Intercept) 0.1704   0.4128
##   site_id      (Intercept) 0.1066   0.3264
## Number of obs: 210, groups: plot_id:site_id, 36; site_id, 12

```

```
##
## Conditional model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.6268    0.5498   1.140 0.254286
## plot_typefield    -1.1086    0.3074  -3.606 0.000311 ***
## bee_season_typer  -2.4574    0.4157  -5.911 3.4e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.6221    0.8838  -0.704   0.481
```

```
##AIC improved slightly with this model.
```

```
#compare 2 models using likelihood ratio test
anova(nsq_glm_zeroPois, nsq_glm_zeroPois_noArea)
```

```
## Data: surveys
## Models:
## nsq_glm_zeroPois_noArea: nest_searching_queens ~ plot_type + bee_season_type + (1 | site_id/plot_id), zi
## nsq_glm_zeroPois: nest_searching_queens ~ plot_type + bee_season_type + (1 | area/site_id/plot_id), zi
##           Df      AIC      BIC logLik deviance Chisq Chi Df
## nsq_glm_zeroPois_noArea  6 297.41 317.50 -142.71  285.41
## nsq_glm_zeroPois        7 299.41 322.84 -142.71  285.41      0      1
##           Pr(>Chisq)
## nsq_glm_zeroPois_noArea
## nsq_glm_zeroPois          1
```

```
##p=1. This tells us that adding area model does not improve model. I should go with model that drops area
```

```
#try model without area and site
nsq_glm_zeroPois_noAreaSite <- glmmTMB(nest_searching_queens ~ plot_type +
  bee_season_type +
  (1|plot_id),
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = surveys)
```

```
#compare 2 models using likelihood ratio test
anova(nsq_glm_zeroPois_noAreaSite, nsq_glm_zeroPois_noArea)
```

```
## Data: surveys
## Models:
## nsq_glm_zeroPois_noAreaSite: nest_searching_queens ~ plot_type + bee_season_type + (1 | plot_id), zi
## nsq_glm_zeroPois_noArea: nest_searching_queens ~ plot_type + bee_season_type + (1 | site_id/plot_id), zi
##           Df      AIC      BIC logLik deviance Chisq Chi Df
## nsq_glm_zeroPois_noAreaSite  5 297.73 314.46 -143.86  287.73
## nsq_glm_zeroPois_noArea      6 297.41 317.50 -142.71  285.41 2.3129      1
##           Pr(>Chisq)
## nsq_glm_zeroPois_noAreaSite
## nsq_glm_zeroPois_noArea          0.1283
```

##p=0.1283. This tells us that adding site does not significantly improve model. I should go with model

```
nsq_glm_zeroPois_noRE <- glmmTMB(nest_searching_queens ~ plot_type +  
                                bee_season_type,  
                                ziformula = ~1, # zero-inflation model  
                                family = poisson, #poisson  
                                data = surveys)
```

#compare 2 models using likelihood ratio test

```
anova(nsq_glm_zeroPois_noArea, nsq_glm_zeroPois_noRE)
```

Data: surveys

Models:

nsq_glm_zeroPois_noRE: nest_searching_queens ~ plot_type + bee_season_type, zi=~1, disp=~1

nsq_glm_zeroPois_noArea: nest_searching_queens ~ plot_type + bee_season_type + (1 | site_id/plot_id)

##	Df	AIC	BIC	logLik	deviance	Chisq	Chi	Df
----	----	-----	-----	--------	----------	-------	-----	----

## nsq_glm_zeroPois_noRE	4	295.73	309.11	-143.86	287.73			
--------------------------	---	--------	--------	---------	--------	--	--	--

## nsq_glm_zeroPois_noArea	6	297.41	317.50	-142.71	285.41	2.3129		2
----------------------------	---	--------	--------	---------	--------	--------	--	---

##	Pr(>Chisq)
----	------------

## nsq_glm_zeroPois_noRE	
--------------------------	--

## nsq_glm_zeroPois_noArea	0.3146
----------------------------	--------

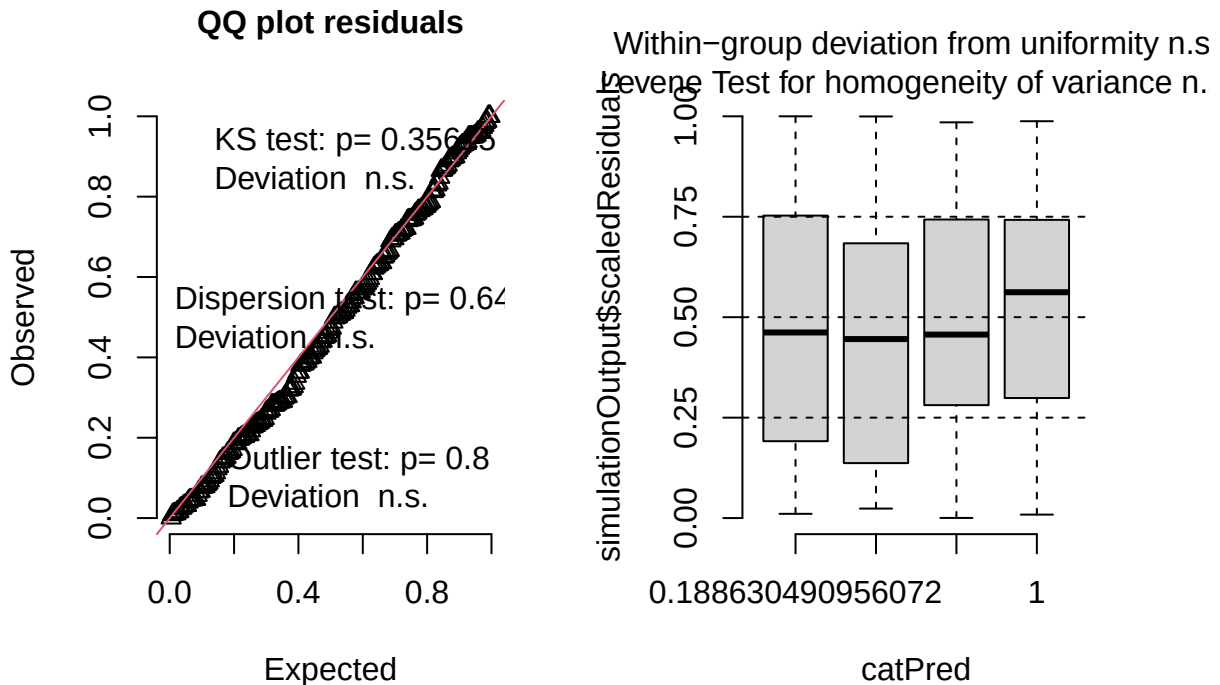
##p=0.e146. This tells us that adding random effects does not improve model. I should go with model wit

Diagnostics

#simulate residuals

```
residuals_nsq_glm_zeroPois_noRE <- simulateResiduals(fittedModel = nsq_glm_zeroPois_noRE, plot = TRUE)
```

DHARMA residual



##no evidence of overdispersion or significant outliers

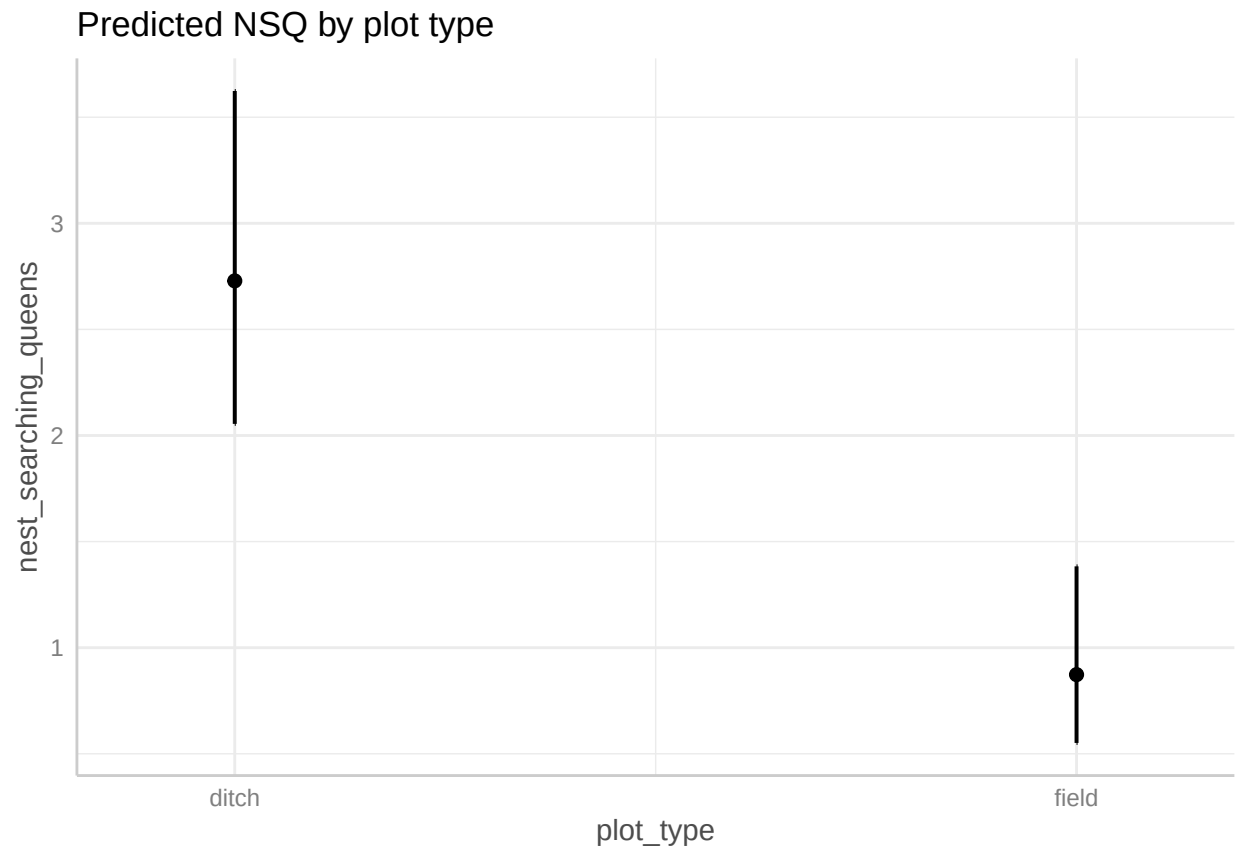
Plot results of model

```
summary(nsq_glm_zeroPois_noRE)
```

```
## Family: poisson ( log )
## Formula:      nest_searching_queens ~ plot_type + bee_season_type
## Zero inflation: ~1
## Data: surveys
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    295.7    309.1   -143.9    287.7      206
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.0038    0.1447   6.936 4.03e-12 ***
## plot_typefield    -1.1396    0.2612  -4.363 1.28e-05 ***
## bee_season_typer  -2.5017    0.4202  -5.954 2.62e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.1598    0.3205  -0.498   0.618
```

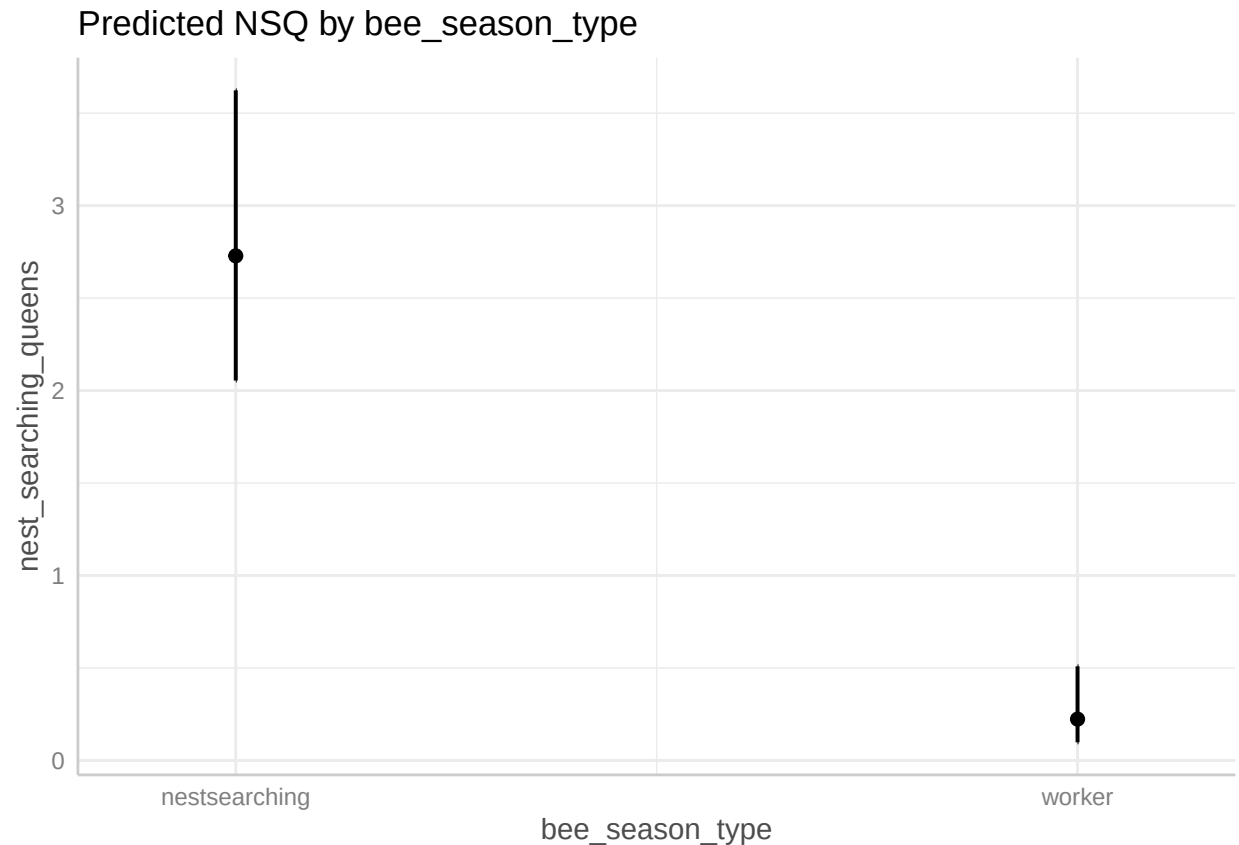
```
##number of nest searching queens vs plot type
# Get predicted values
pred1 <- ggpredict(nsq_glm_zeroPois_noRE, terms = "plot_type")

# Plot
plot(pred1) + ggtitle("Predicted NSQ by plot type")
```



```
##number of nest searching queens vs season
# Get predicted values
pred2 <- ggpredict(nsq_glm_zeroPois_noRE, terms = "bee_season_type")

# Plot
plot(pred2) + ggtitle("Predicted NSQ by bee_season_type")
```

GLM #2: Number of NSQ vs veg variables

Choosing which vegetation variables to include as predictors by checking correlation among the variables

```
vif(lm(nest_searching_queens ~ bare_perc_scaled +
      grass_forb_perc_scaled +
      grass_h_scaled +
      forb_h_scaled +
      bombus_floral_perc_scaled +
      flood_perc_scaled +
      dry_root_perc_scaled +
      bombus_floral_richness_scaled +
      bombus_floral_abun_scaled + bee_season_type,
      data = surveys))
```

```
##          bare_perc_scaled      grass_forb_perc_scaled
##          1.619483          2.833229
##          grass_h_scaled      forb_h_scaled
##          4.200729          4.105429
## bombus_floral_perc_scaled      flood_perc_scaled
##          1.176476          1.104416
##      dry_root_perc_scaled bombus_floral_richness_scaled
##          1.280929          2.589861
## bombus_floral_abun_scaled      bee_season_type
```

```
##                                1.429439                                1.398747
```

```
##all variables have VIF <5 (no multicollinearity). so i can include all these predictors in my model.
```

Run PCA to determine which vegetation variables to include

```
pca_res <- prcomp(surveys[, c("bare_perc_scaled",
                              "grass_forb_perc_scaled",
                              "grass_h_scaled",
                              "forb_h_scaled",
                              "bombus_floral_perc_scaled",
                              "flood_perc_scaled",
                              "dry_root_perc_scaled",
                              "bombus_floral_richness_scaled",
                              "bombus_floral_abun_scaled")], scale. = TRUE)

summary(pca_res)
```

```
## Importance of components:
```

```
##              PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation    1.8069 1.2146 1.0893 0.94377 0.89920 0.81832 0.54103
## Proportion of Variance 0.3628 0.1639 0.1318 0.09897 0.08984 0.07441 0.03252
## Cumulative Proportion 0.3628 0.5267 0.6585 0.75747 0.84731 0.92171 0.95423
##              PC8      PC9
## Standard deviation    0.51475 0.38330
## Proportion of Variance 0.02944 0.01632
## Cumulative Proportion 0.98368 1.00000
```

```
#see which variable drives each of the principle components
```

```
print(pca_res$rotation)
```

```
##              PC1      PC2      PC3      PC4
## bare_perc_scaled    0.23888893 0.514444969 0.14361858 -0.5052882159
## grass_forb_perc_scaled -0.42153457 -0.22667141 -0.29876891 -0.0004167371
## grass_h_scaled      -0.49631286 -0.11630289 -0.04835346 -0.1513395584
## forb_h_scaled       -0.48406660 -0.08870371 0.11125744 -0.2446042385
## bombus_floral_perc_scaled -0.19026245 0.40159218 -0.16645038 0.6561223125
## flood_perc_scaled    0.12371312 -0.08420063 -0.63744066 0.0122999251
## dry_root_perc_scaled  0.02201065 -0.39645174 0.62688408 0.3126314992
## bombus_floral_richness_scaled -0.43122902 0.25793301 0.13996816 -0.2391583431
## bombus_floral_abun_scaled -0.21597706 0.52009367 0.16965703 0.2763436361
##              PC5      PC6      PC7      PC8
## bare_perc_scaled    0.07533404 0.36825434 0.50474263 0.062418976
## grass_forb_perc_scaled -0.28434372 -0.14493023 0.58996688 0.439577371
## grass_h_scaled      0.03083221 0.21511901 0.17032135 -0.396955743
## forb_h_scaled       0.15796482 0.17778710 -0.12501627 -0.405217154
## bombus_floral_perc_scaled -0.09494476 0.57132504 -0.03084109 0.046523209
## flood_perc_scaled    0.74980395 0.02580597 0.08680372 0.007530054
## dry_root_perc_scaled  0.43638944 0.18560622 0.30656671 0.180555048
## bombus_floral_richness_scaled 0.27204694 -0.01848605 -0.42808270 0.625266871
## bombus_floral_abun_scaled 0.22782488 -0.63531872 0.25847927 -0.235203967
##              PC9
## bare_perc_scaled    -0.04882006
```

```
## grass_forb_perc_scaled      -0.19626601
## grass_h_scaled              0.69361043
## forb_h_scaled               -0.67023174
## bombus_floral_perc_scaled   -0.07602182
## flood_perc_scaled           -0.02568914
## dry_root_perc_scaled        0.01227843
## bombus_floral_richness_scaled 0.14886089
## bombus_floral_abun_scaled   -0.00637804
```

PCA Results:

PC1: grass_h_scaled (-0.50), forb_h_scaled (-0.48), grass_forb_perc_scaled (-0.42), bombus_floral_richness_scaled (-0.43), bombus_floral_abun_scaled (-0.22)

PC2: bare_perc_scaled (0.51), bombus_floral_abun_scaled (0.52), bombus_floral_perc_scaled (0.40), dry_root_perc_scaled (-0.40)

PC3: dry_root_perc_scaled (0.63), flood_perc_scaled (-0.64)

Based on these PCA results, I am selecting to include the following predictors in my model:

- grass_forb_perc_scaled (loads strongly on PCA1)
- bombus_floral_abun_scaled (loads strongly on PCA1 & 2)
- flood_perc_scaled (loads strongly on PCA3)

Check pairwise correlations between this reduced set of variables

```
cor(surveys[, c("grass_forb_perc_scaled", "flood_perc_scaled", "bombus_floral_abun_scaled")])
```

```
##               grass_forb_perc_scaled flood_perc_scaled
## grass_forb_perc_scaled              1.00000000      -0.07441975
## flood_perc_scaled                  -0.07441975       1.00000000
## bombus_floral_abun_scaled           0.08976466      -0.14386003
##               bombus_floral_abun_scaled
## grass_forb_perc_scaled              0.08976466
## flood_perc_scaled                  -0.14386003
## bombus_floral_abun_scaled           1.00000000
```

##correlation among all the pairs are all near zero --> predictors are all sufficiently independent!

DITCH

Filter to only include ditch data

```
ditch_data <- surveys %>%
  filter(plot_id == "d")
```

Add random effects and check which is best

```

nsq_ditch_zeroPois <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                             bombus_floral_abun_scaled +
                             flood_perc_scaled +
                             bee_season_type +
                             (1|area/site_id/plot_id),
                             ziformula = ~1, # zero-inflation model
                             family = poisson, #poisson
                             data = ditch_data)

```

```

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')

```

#model convergence problem

```

diagnose(nsq_ditch_zeroPois)

```

```

## Unusually large coefficients (|x|>10):
##
##      (Intercept) flood_perc_scaled      zi~(Intercept)
##      -34.47558      -121.09882      -47.24270
##
## Large negative coefficients in zi (log-odds of zero-inflation),
## dispersion, or random effects (log-standard deviations) suggest
## unnecessary components (converging to zero on the constrained scale);
## large negative and/or positive components in binomial or Poisson
## conditional parameters suggest (quasi-)complete separation. Large
## values of nbinom2 dispersion suggest that you should use a Poisson
## model instead.

```

##tells us that zero-inflation term may not be necessary.

```

nsq_ditch_Pois <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                          bombus_floral_abun_scaled +
                          flood_perc_scaled +
                          bee_season_type +
                          (1|area/site_id/plot_id),
                          family = poisson, #poisson
                          data = ditch_data)

```

```

## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; singular convergence (7). See vignette('troubleshooting'),
## help('diagnose')

```

```

diagnose(nsq_ditch_Pois)

```

```

## Unusually large coefficients (|x|>10):
##
##      (Intercept) flood_perc_scaled
##      -34.18934      -120.09202
##

```

```
## Large negative coefficients in zi (log-odds of zero-inflation),
## dispersion, or random effects (log-standard deviations) suggest
## unnecessary components (converging to zero on the constrained scale);
## large negative and/or positive components in binomial or Poisson
## conditional parameters suggest (quasi-)complete separation. Large
## values of nbinom2 dispersion suggest that you should use a Poisson
## model instead.
```

##try dropping flood perc.. makes sense since there isn't flooding along the ditch. Only within the fie

```
nsq_ditch_Pois <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                        bombus_floral_abun_scaled +
                        bee_season_type +
                        (1|area/site_id/plot_id),
                        family = poisson, #poisson
                        data = ditch_data)

#test different random effect structures

nsq_ditch_Pois_noArea <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                                bombus_floral_abun_scaled +
                                bee_season_type +
                                (1|site_id/plot_id),
                                family = poisson, #poisson
                                data = ditch_data)

anova(nsq_ditch_Pois, nsq_ditch_Pois_noArea)
```

```
## Data: ditch_data
## Models:
## nsq_ditch_Pois_noArea: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_ditch_Pois_noArea:      bee_season_type + (1 | site_id/plot_id), zi=~0, disp=~1
## nsq_ditch_Pois: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , zi=~0
## nsq_ditch_Pois:      bee_season_type + (1 | area/site_id/plot_id), zi=~0, disp=~1
##
##      Df    AIC    BIC logLik deviance Chisq Chi Df
## nsq_ditch_Pois_noArea  6 144.46 157.95 -66.231   132.46
## nsq_ditch_Pois        7 145.81 161.55 -65.907   131.81 0.6476      1
##
##      Pr(>Chisq)
## nsq_ditch_Pois_noArea
## nsq_ditch_Pois      0.421
```

##more complex model doesn't have better fit

```
nsq_ditch_Pois_noAreaSite <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                                    bombus_floral_abun_scaled +
                                    bee_season_type +
                                    (1|plot_id),
                                    family = poisson, #poisson
                                    data = ditch_data)
anova(nsq_ditch_Pois_noArea, nsq_ditch_Pois_noAreaSite)
```

```
## Data: ditch_data
```

```
## Models:
## nsq_ditch_Pois_noAreaSite: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_ditch_Pois_noAreaSite:      bee_season_type + (1 | plot_id), zi=~0, disp=~1
## nsq_ditch_Pois_noArea: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_ditch_Pois_noArea:      bee_season_type + (1 | site_id/plot_id), zi=~0, disp=~1
##
##           Df      AIC      BIC logLik deviance Chisq Chi Df
## nsq_ditch_Pois_noAreaSite  5 159.75 171.00 -74.877   149.75
## nsq_ditch_Pois_noArea      6 144.46 157.95 -66.231   132.46 17.291      1
##
##           Pr(>Chisq)
## nsq_ditch_Pois_noAreaSite
## nsq_ditch_Pois_noArea      3.206e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

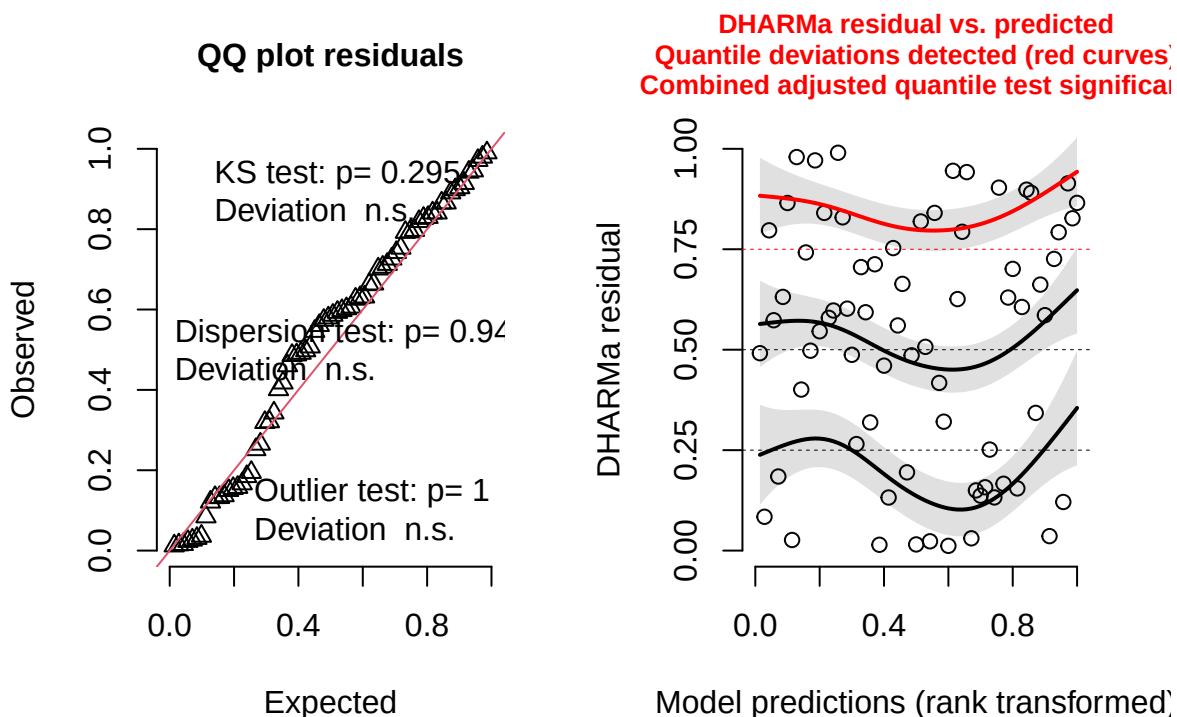
##more complex model has better fit. go with plot nested in site

Diagnostics

#simulate residuals

```
residuals_nsq_ditch_Pois_noArea <- simulateResiduals(fittedModel = nsq_ditch_Pois_noArea, plot = TRUE)
```

DHARMA residual



##there is no evidence of overdispersion or significant outliers

Plot results of model

```
summary(nsq_ditch_Pois_noArea)
```

```
## Family: poisson ( log )
## Formula:
## nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
##     bee_season_type + (1 | site_id/plot_id)
## Data: ditch_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    144.5    158.0    -66.2    132.5      64
##
## Random effects:
##
## Conditional model:
## Groups          Name      Variance Std.Dev.
## plot_id:site_id (Intercept) 0.4855  0.6968
## site_id         (Intercept) 0.4855  0.6968
## Number of obs: 70, groups: plot_id:site_id, 12; site_id, 12
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.2374    0.4034  -0.589   0.5562
## grass_forb_perc_scaled    0.1886    0.2268   0.831   0.4058
## bombus_floral_abun_scaled -0.2335    0.1347  -1.733   0.0831 .
## bee_season_typeworker    -2.4165    0.5352  -4.515 6.32e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

##only effect of season is significant. marginal negative effect of floral abundance on nsq

P0-lotting a binary model to predict if #NSQ > 0

```
#create binary variable for nsq presence
ditch_data$nsq_binary <- ifelse(ditch_data$nest_searching_queens > 0, 1, 0)

nsq_ditch_binary_allRE <- glmmTMB(nsq_binary ~ grass_forb_perc_scaled +
                                bombus_floral_abun_scaled +
                                bee_season_type +
                                (1|area/site_id/plot_id),
                                family = binomial,
                                data = ditch_data)

nsq_ditch_binary_noArea <- glmmTMB(nsq_binary ~ grass_forb_perc_scaled +
                                bombus_floral_abun_scaled +
                                bee_season_type +
                                (1|site_id/plot_id),
                                family = binomial,
                                data = ditch_data)
```

```
## Warning in finalizeTMB(TMBStruc, obj, fit, h, data.tmb.old): Model convergence
## problem; non-positive-definite Hessian matrix. See vignette('troubleshooting')
```

```
#no convergence
```

```
nsq_ditch_binary_noAreaSite <- glmmTMB(nsq_binary ~ grass_forb_perc_scaled +  
  bombus_floral_abun_scaled +  
  bee_season_type +  
  (1|plot_id),  
  family = binomial,  
  data = ditch_data)
```

```
nsq_ditch_binary_noRE <- glmmTMB(nsq_binary ~ grass_forb_perc_scaled +  
  bombus_floral_abun_scaled +  
  bee_season_type,  
  family = binomial,  
  data = ditch_data)
```

```
AIC(nsq_ditch_binary_allRE,  
  nsq_ditch_binary_noAreaSite,  
  nsq_ditch_binary_noRE  
)
```

```
##              df      AIC  
## nsq_ditch_binary_allRE      7 78.12939  
## nsq_ditch_binary_noAreaSite  5 76.32430  
## nsq_ditch_binary_noRE       4 74.32430
```

```
##model with no random effects is best
```

```
summary(nsq_ditch_binary_noRE)
```

```
## Family: binomial ( logit )  
## Formula:  
## nsq_binary ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +  
## bee_season_type  
## Data: ditch_data  
##  
##      AIC      BIC    logLik -2*log(L)  df.resid  
##      74.3      83.3     -33.2      66.3       66  
##  
##  
## Conditional model:  
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)      0.00811    0.41710   0.019 0.984487  
## grass_forb_perc_scaled  0.24128    0.30079   0.802 0.422460  
## bombus_floral_abun_scaled -0.23485    0.34502  -0.681 0.496075  
## bee_season_typeworker   -2.48938    0.73117  -3.405 0.000662 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##no significant predictors other than season
```

Field veg variables

Filter to only include ditch data


```
field_data <- surveys %>%
  filter(plot_id != "d")
```

Plotting a zero-inflated poisson model (Q: should I check random effects for this model too?)

```
nsq_field_zeroPois <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
  bombus_floral_abun_scaled +
  flood_perc_scaled +
  bee_season_type +
  (1|area/site_id/plot_id),
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = field_data)

#test different random effect structures
nsq_field_zeroPois_noArea <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
  bombus_floral_abun_scaled +
  flood_perc_scaled +
  bee_season_type +
  (1|site_id/plot_id),
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = field_data)

anova(nsq_field_zeroPois, nsq_field_zeroPois_noArea)
```

```
## Data: field_data
## Models:
## nsq_field_zeroPois_noArea: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled
## nsq_field_zeroPois_noArea:      flood_perc_scaled + bee_season_type + (1 | site_id/plot_id), zi=~1, d
## nsq_field_zeroPois: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled + , z
## nsq_field_zeroPois:      flood_perc_scaled + bee_season_type + (1 | area/site_id/plot_id), zi=~1, disp
##
##          Df      AIC      BIC    logLik deviance Chisq Chi Df
## nsq_field_zeroPois_noArea  8 158.75 182.28 -71.375    142.75
## nsq_field_zeroPois        9 160.75 187.22 -71.375    142.75      0      1
##
##          Pr(>Chisq)
## nsq_field_zeroPois_noArea
## nsq_field_zeroPois          0.9999
```

```
##more complex model doesn't have better fit

nsq_field_zeroPois_noAreaSite <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
  bombus_floral_abun_scaled +
  flood_perc_scaled +
  bee_season_type +
  (1|plot_id),
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = field_data)

anova(nsq_field_zeroPois_noArea, nsq_field_zeroPois_noAreaSite)
```

```
## Data: field_data
## Models:
```

```
## nsq_field_zeroPois_noAreaSite: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_field_zeroPois_noAreaSite:      flood_perc_scaled + bee_season_type + (1 | plot_id), zi=~1, disp=1
## nsq_field_zeroPois_noArea: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_field_zeroPois_noArea:      flood_perc_scaled + bee_season_type + (1 | site_id/plot_id), zi=~1, disp=1
##
##      Df      AIC      BIC logLik deviance Chisq Chi Df
## nsq_field_zeroPois_noAreaSite  7 156.51 177.10 -71.254   142.51
## nsq_field_zeroPois_noArea     8 158.75 182.28 -71.375   142.75      0      1
##
##      Pr(>Chisq)
## nsq_field_zeroPois_noAreaSite
## nsq_field_zeroPois_noArea      1
```

##more complex model doesn't have better fit

```
nsq_field_zeroPois_noRE <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
                                bombus_floral_abun_scaled +
                                flood_perc_scaled +
                                bee_season_type ,
                                ziformula = ~1, # zero-inflation model
                                family = poisson, #poisson
                                data = field_data)
anova(nsq_field_zeroPois_noAreaSite, nsq_field_zeroPois_noRE)
```

```
## Data: field_data
## Models:
## nsq_field_zeroPois_noRE: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_field_zeroPois_noRE:      flood_perc_scaled + bee_season_type, zi=~1, disp=1
## nsq_field_zeroPois_noAreaSite: nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## nsq_field_zeroPois_noAreaSite:      flood_perc_scaled + bee_season_type + (1 | plot_id), zi=~1, disp=1
##
##      Df      AIC      BIC logLik deviance Chisq Chi Df
## nsq_field_zeroPois_noRE      6 154.75 172.4 -71.375   142.75
## nsq_field_zeroPois_noAreaSite  7 156.51 177.1 -71.254   142.51 0.2404      1
##
##      Pr(>Chisq)
## nsq_field_zeroPois_noRE
## nsq_field_zeroPois_noAreaSite  0.6239
```

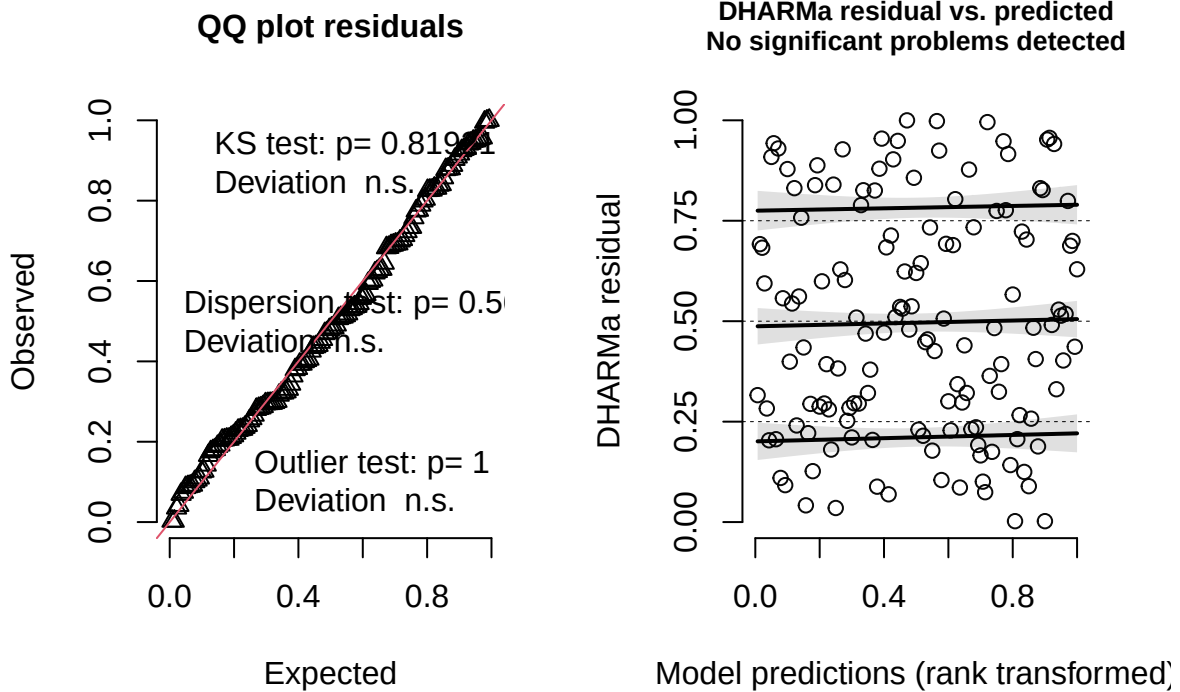
##more complex model doesn't have better fit.

Diagnostics

#simulate residuals

```
residuals_nsq_field_zeroPois_noRE <- simulateResiduals(fittedModel = nsq_field_zeroPois_noRE, plot = TRUE)
```

DHARMa residual



##no evidence of overdispersion or significant outliers

Plot results of model

`summary(nsqr_field_zeroPois_noRE)`

```
## Family: poisson ( log )
## Formula:
## nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
##      flood_perc_scaled + bee_season_type
## Zero inflation: ~1
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    154.7    172.4    -71.4    142.7      134
##
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.35651    0.31528  -1.131  0.25815
## grass_forb_perc_scaled  0.36991    0.22184   1.667  0.09542 .
## bombus_floral_abun_scaled 0.29295    0.20776   1.410  0.15854
## flood_perc_scaled   -0.08541    0.19679  -0.434  0.66430
## bee_season_typerworker -2.24975    0.63059  -3.568  0.00036 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.4081    0.7007  -0.583    0.56
```

```
##marginal positive effect of grass+forb
```

Plotting a binary model to predict if #NSQ > 0

```
#create binary variable for nsq presence
field_data$nsq_binary <- ifelse(field_data$nest_searching_queens > 0, 1, 0)

nsq_field_binary <- glmmTMB(nsq_binary ~ grass_forb_perc_scaled +
                           bombus_floral_abun_scaled +
                           flood_perc_scaled +
                           bee_season_type,
                           family = binomial,
                           data = field_data)

summary(nsq_field_binary)
```

```
## Family: binomial ( logit )
## Formula:
## nsq_binary ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
## flood_perc_scaled + bee_season_type
## Data: field_data
##
##           AIC          BIC    logLik -2*log(L)  df.resid
##           99.3          114.0    -44.7      89.3      135
##
##
## Conditional model:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.8221    0.3024  -2.718 0.006564 **
## grass_forb_perc_scaled    0.7684    0.3153   2.437 0.014812 *
## bombus_floral_abun_scaled    0.3936    0.2843   1.385 0.166150
## flood_perc_scaled    -0.3355    0.3035  -1.105 0.268982
## bee_season_typerworker    -2.7269    0.7830  -3.483 0.000497 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##significant positive effect of grass+forb cover on nesting abundance
```